

PipelineIQ

Data Quality & Pipeline Intelligence Platform

Recommended Resume Title

Founder | Data Platform Engineer

Alternative Titles:

- Founder | Data Engineering Consultant
 - Founder | Analytics Infrastructure Engineer
 - Founder | ETL Systems Engineer
 - Founder | Data Reliability Engineer
 - Founder | Data Quality Engineer
 - Founder | Data Operations Engineer
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Executive Summary

PipelineIQ is a cloud-based data quality and pipeline intelligence platform that helps organizations validate, monitor, and improve the quality of their data before it enters analytics, reporting, machine learning, or operational systems.

Organizations constantly struggle with:

- Missing records
- Duplicate data
- Invalid values
- Broken imports
- Schema inconsistencies
- Data drift
- Reporting errors

Many organizations don't discover these problems until after dashboards break or business decisions are made using incorrect information.

PipelineIQ solves this problem by allowing users to upload datasets and automatically receive data quality assessments, anomaly detection reports, schema validation results, and actionable remediation recommendations.

The platform functions as a lightweight version of products used by modern data engineering teams.

Think:

- DataDog Data Observability
- Monte Carlo Data
- Great Expectations
- Soda
- Bigeye

but built entirely using free technologies.

Why This Project Matters

Most data engineering portfolios demonstrate:

- ETL scripts
- SQL queries
- Data warehouses

Very few demonstrate:

- Data reliability
- Data observability
- Quality assurance automation
- Pipeline intelligence

Those are some of the fastest-growing areas in modern data engineering.

PipelineIQ allows Allyson to showcase:

Backend Engineering

- API design
- Data processing

Data Engineering

- ETL workflows
- Validation frameworks

Analytics Engineering

- Data quality monitoring
- Business intelligence support

Cloud Engineering

- Deployment
- Scalable processing

Platform Engineering

- User-facing data products

Target Users

Data Analysts

Business Intelligence Teams

Nonprofits

Healthcare Organizations

Small Businesses

Data Engineers

Operations Teams

Researchers

Technology Stack

Everything is free.

Frontend

- React

Backend

- Python
- FastAPI

Database

- PostgreSQL

Data Processing

- Pandas
- NumPy

Validation Engine

- Great Expectations (Open Source)

Visualization

- Plotly

Deployment

- Render

Source Control

- GitHub

Containerization

- Docker

Development Tools

- VS Code

Total Cost: \$0

Core Product Workflow

User uploads:

CSV

Excel

JSON

Database Export

The platform automatically performs:

Schema Validation

Missing Value Analysis

Duplicate Detection

Null Pattern Analysis

Outlier Detection

Data Drift Detection

Quality Scoring

Recommendations

Platform Modules

Module 1

Data Intake Center

Purpose

Receive uploaded datasets.

Supported Formats

CSV

Excel

JSON

Features

Upload Interface

File Preview

Column Inspection

Metadata Extraction

Deliverable

Data ingestion system.

Module 2

Schema Validation Engine

Purpose

Verify dataset structure.

Detects

Missing Columns

Unexpected Columns

Incorrect Data Types

Invalid Formats

Example

Expected:

Date Column

Received:

Text Column

Validation Failure Generated.

Deliverable

Schema validation framework.

Module 3

Data Quality Intelligence Engine

Purpose

Identify quality issues.

Metrics

Missing Values

Duplicate Records

Invalid Values

Data Consistency

Completeness

Accuracy

Example

Customer ID

Missing:

14%

Quality Warning Generated.

Deliverable

Quality assessment engine.

Module 4

Data Health Score

Purpose

Create a single quality indicator.

Example Formula

Completeness

Consistency

Uniqueness

Validity

Timeliness

Combined Into:

Data Health Score

0–100

Example

Dataset Score:

87/100

Deliverable

Automated scoring system.

Module 5

Pipeline Readiness Assessment

Purpose

Determine whether data is suitable for analytics.

Categories

Ready

Minor Issues

Needs Review

Critical Failure

Example

Pipeline Readiness

73%

Recommendation:

Fix duplicate records before import.

Deliverable

Readiness evaluation engine.

Module 6

Anomaly Detection Center

Purpose

Identify unusual data patterns.

Examples

Revenue spikes

Missing transactions

Unexpected volume changes

Outlier detection

Example

Typical Range:

100-200 records/day

Current Upload:

2,500 records/day

Anomaly Detected.

Deliverable

Anomaly detection module.

Module 7

Recommendation Engine

Purpose

Provide actionable fixes.

Recommendations

Remove duplicates

Standardize dates

Correct null values

Validate reference tables

Normalize categories

Example

Issue:

Customer IDs duplicated.

Recommendation:

Create uniqueness constraint.

Deliverable

Automated remediation suggestions.

Module 8

Executive Reporting Center

Purpose

Generate professional reports.

Outputs

Data Quality Report

Validation Report

Pipeline Readiness Report

Executive Summary

Export Formats

PDF

CSV
JSON

Deliverable

Reporting system.

Public Data Demonstration Layer

The platform should work with user-uploaded datasets.

However, Allyson should include publicly available demonstration datasets so recruiters can immediately test the platform.

Examples:

Population Data

Available from:

United States Census Bureau

Labor Statistics

Available from:

Bureau of Labor Statistics

Healthcare Data

Available from:

Centers for Medicare & Medicaid Services

Open Government Data

Available from:

Data.gov

These datasets are only included for demonstration.

The primary platform value comes from user-uploaded data.

Week 1

Foundation & Architecture

Objectives

Build platform architecture.

Tasks

Install:

- Python
- FastAPI
- PostgreSQL
- Docker

Create:

- GitHub repository

Design:

- Database schema

Create:

- API structure

Deploy:

- Initial Render application

Deliverable

Working cloud deployment.

Week 2

Data Ingestion Engine

Objectives

Build upload system.

Tasks

Create:

- Upload workflows

Implement:

- CSV processing

Implement:

- Excel processing

Create:

- Metadata extraction

Store:

- Dataset profiles

Deliverable

File upload system.

Week 3

Validation Framework

Objectives

Build quality checks.

Tasks

Implement:

- Schema validation

Create:

- Missing value analysis

Build:

- Duplicate detection

Create:

- Data type validation

Deliverable

Quality validation engine.

Week 4

Scoring & Analytics

Objectives

Generate intelligence.

Tasks

Create:

- Data Health Score

Develop:

- Pipeline Readiness Score

Implement:

- Anomaly detection

Create:

- Recommendation engine

Deliverable

Intelligence engine.

Week 5

Reporting & Visualization

Objectives

Build executive outputs.

Tasks

Create:

- Dashboards

Build:

- Quality reports

Implement:

- PDF exports

Create:

- Interactive visualizations

Deliverable

Reporting center.

Week 6

Production Release

Objectives

Portfolio-ready launch.

Tasks

Improve:

- UI/UX

Write:

- Documentation

Create:

- Architecture diagrams

Add:

- Demo datasets

Record:

- Walkthrough video

Deploy:

- Final version

Deliverable

Public release.

Final Deliverables

At completion Allyson will have:

Public URL

Example:

pipelineiq.onrender.com

GitHub Repository

Production-quality source code.

Technical Documentation

Architecture

API Design

Data Models

Validation Logic

Demo Datasets

Publicly available examples.

Walkthrough Video

Recruiter-friendly demonstration.

Architecture Diagram

Enterprise-style design document.

Resume Bullets

- Designed and deployed PipelineIQ, a cloud-based data quality and pipeline intelligence platform that validates uploaded datasets and generates automated data reliability assessments.
- Engineered a scalable FastAPI and PostgreSQL backend supporting schema validation, anomaly detection, duplicate identification, and automated quality scoring workflows.
- Developed ETL-style ingestion pipelines capable of processing CSV, Excel, JSON, and database export files while generating metadata profiles and validation reports.
- Built automated Data Health Score and Pipeline Readiness Score frameworks that evaluate dataset quality, consistency, completeness, and operational reliability.
- Implemented interactive analytics dashboards and executive reporting capabilities using Python, Plotly, and React to support data governance and decision-making.
- Deployed a production-ready platform using Docker, GitHub, and Render, demonstrating expertise in data engineering, backend systems, cloud deployment, and data observability.